

Class 22 Fermat's Last Theorem

Goals

- 1 Know the statement of Fermat's Last Theorem and its similarity to the Pythagorean Theorem
2. Use Fermat's Last Theorem to prove some given number is irrational.

We saw last class that the equation $x^2 + y^2 = z^2$ is Diophantine since it has integer solutions. For centuries it was undetermined if this were true for equations with integer exponents greater than 2.

Prove that $\sqrt[3]{9}$ is irrational.

Proof 1: Use Fermat's Last Theorem.

Proof 2: Use Euclid's Lemma